

WEEKEND PRACTICAL ASSIGNMENT

**BUILD AND CONFIGURE A SECURE LINUX SERVER
FOR A STARTUP COMPANY**

ASSIGNMENT - 1 (WEEK 1)

**SUBMITTED AS PART OF THE
MASTER CLASS PROGRAM ON CLOUD COMPUTING AND DEVOPS**

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Introduction

Project Scenario

ABC Tech Solutions Pvt. Ltd. is setting up a new Linux server to be used by three internal teams - Developers, Testers, and Managers. As a Junior Linux and DevOps Engineer, the objective of this assignment was to configure, secure, and prepare this server for production-style use. This included creating and organizing user accounts, restricting access through directory permissions, enabling secure remote administration, hosting a company webpage, and automating routine administrative checks through Bash scripting.

Objectives

- Configure and manage Linux users and groups.
- Implement secure file and directory permissions.
- Enable secure remote administration using SSH.
- Transfer files securely using FTP.
- Host a simple website using the Apache Web Server.
- Develop a Bash script to automate routine administrative tasks.
- Document the work professionally with screenshots and explanations.

Environment Setup

The server was set up as an Ubuntu Linux virtual machine running on Oracle VirtualBox. The system hostname is yashdeep-kaur-VirtualBox and the internal network IP address assigned to the VM is 10.0.2.15. All administrative tasks described in this report were carried out directly on this VM using a terminal session.

As only a single virtual machine was available for this assignment, tasks that require a client-server demonstration (SSH login, FTP transfer) were carried out by connecting to the server's own network IP address (10.0.2.15) rather than from a separate physical machine. This still fully exercises the authentication and network service pathways being tested.

Task 1: User and Group Management

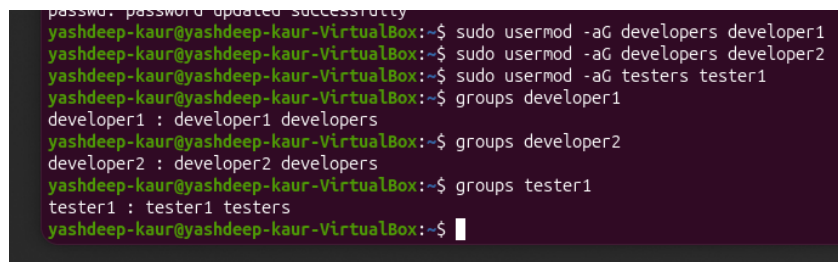
Objective: Create user accounts for developers, testers, and a manager, and organize them into appropriate groups.

1.1 Commands Executed

```
sudo groupadd developers
sudo groupadd testers
sudo useradd -m developer1
sudo useradd -m developer2
sudo useradd -m tester1
sudo useradd -m manager
sudo passwd developer1(repeated for each user)
```

```
sudo usermod -aG developers developer1
sudo usermod -aG developers developer2
sudo usermod -aG testers tester1
groups developer1 groups developer2 groups tester1
```

1.2 Result



```
password: password updated successfully
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ sudo usermod -aG developers developer1
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ sudo usermod -aG developers developer2
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ sudo usermod -aG testers tester1
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ groups developer1
developer1 : developer1 developers
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ groups developer2
developer2 : developer2 developers
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ groups tester1
tester1 : tester1 testers
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$
```

Figure 1.1 - Verification of user-group assignment (developer1, developer2, tester1).

All four user accounts were created successfully, and group membership was verified using the groups command. developer1 and developer2 were confirmed under the developers group, and tester1 under the testers group.

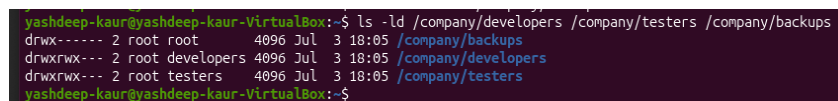
Task 2: Directory Structure and Permissions

Objective: Create a directory hierarchy for the company and configure permissions so each team can only access its own directory, with the backup directory restricted to the administrator.

2.1 Commands Executed

```
sudo mkdir -p /company/developers /company/testers /company/backups
sudo chgrp developers /company/developers
sudo chgrp testers /company/testers
sudo chmod 770 /company/developers
sudo chmod 770 /company/testers
sudo chmod 700 /company/backups
ls -ld /company/developers /company/testers /company/backups
```

2.2 Result



```
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ls -ld /company/developers /company/testers /company/backups
drwx----- 2 root root      4096 Jul  3 18:05 /company/backups
drwxrwx--- 2 root developers 4096 Jul  3 18:05 /company/developers
drwxrwx--- 2 root testers   4096 Jul  3 18:05 /company/testers
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$
```

Figure 2.1 - Final directory permissions on /company (developers, testers, backups).

2.3 Access Control Verification

To confirm the permission boundaries actually work, developer1 was used to test access to both the developers and testers directories.

su - developer1 whoami	developer1 cd
/company/developers pwd	succeeds ->
/company/developers cd /company/testers	fails: Permission denied
exit	

```

$ whoami
developer1
$ su - developer1
Password:
$ cd /company/developers
$ pwd
/company/developers
$ cd /company/testers
-sh: 3: cd: can't cd to /company/testers
$

```

Figure 2.2 - Access control test - developer1 permitted and denied access correctly.

developer1 was able to access /company/developers but was correctly denied access to /company/testers, confirming that the group-based permission structure is functioning as intended.

Task 3: SSH Configuration

Objective: Enable secure remote administration using SSH and demonstrate a successful remote login.

3.1 Commands Executed

```

sudo apt install openssh-server -y
sudo systemctl enable
ssh sudo systemctl start ssh
sudo systemctl status
ssh hostname -I

```

3.2 Result

```

yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 19:26:43 IST; 10min ago
     Invocation: 24372437122c4e7bb71573f5a960f667
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1329 (sshd)
      Tasks: 1 (limit: 1917)
     Memory: 1.1M (peak: 2.3M, swap: 140K, swap peak: 140K)
        CPU: 67ms
     CGroup: /system.slice/ssh.service
            └─1329 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 03 19:26:43 yashdeep-kaur-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 03 19:26:43 yashdeep-kaur-VirtualBox sshd[1329]: Server listening on 0.0.0.0 port 22.
Jul 03 19:26:43 yashdeep-kaur-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jul 03 19:26:43 yashdeep-kaur-VirtualBox sshd[1329]: Server listening on :: port 22.
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$

```

Figure 3.1 - SSH service status - active (running).

```

yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ hostname -I
10.0.2.15 fd17:625c:f037:2:a00:27ff:fe7b:4695
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$

```

Figure 3.2 - Server IP address (10.0.2.15).

3.3 Remote Login Test

```
ssh developer1@10.0.2.15
```

```

yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ssh developer1@10.0.2.15
developer1@10.0.2.15's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-27-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

9 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```

Figure 3.3 - Successful SSH remote login as developer1.

```

$ whoami
developer1
$ hostname
yashdeep-kaur-VirtualBox
$ exit
Connection to 10.0.2.15 closed.
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$

```

Figure 3.4 - Session confirmation (whoami, hostname) and clean exit.

Table 5.1: SSH Session Details

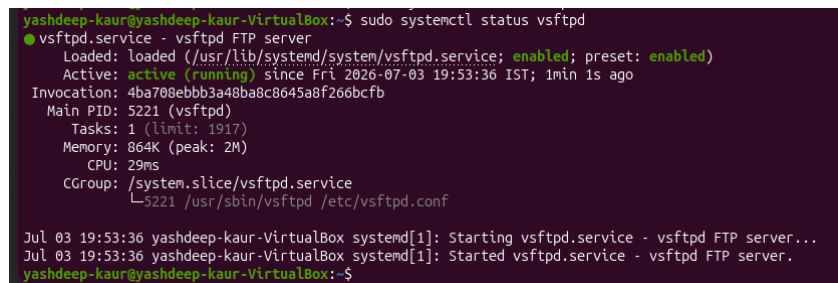
Detail	Value
Server IP Address	10.0.2.15
Login Username	developer1

Task 4: FTP Configuration

Objective: Configure FTP and demonstrate secure file transfer between systems.

4.1 Commands Executed

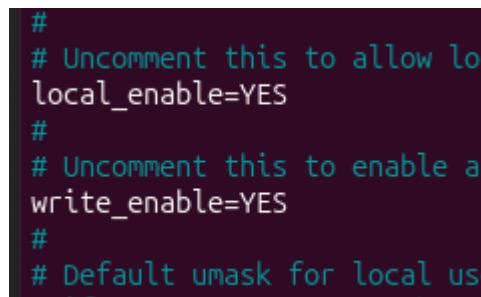
```
sudo apt install vsftpd -y
sudo systemctl enable vsftpd
sudo systemctl start vsftpd
sudo systemctl status vsftpd
sudo nano /etc/vsftpd.conf      local_enable=YES      write_enable=YES
sudo systemctl restart vsftpd
```



```
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 19:53:36 IST; 1min 1s ago
     Invocation: 4ba709ebbb3a48ba8c8645a8f266bcfb
       Main PID: 5221 (vsftpd)
         Tasks: 1 (limit: 1917)
        Memory: 864K (peak: 2M)
           CPU: 29ms
      CGroup: /system.slice/vsftpd.service
              └─5221 /usr/sbin/vsftpd /etc/vsftpd.conf

Jul 03 19:53:36 yashdeep-kaur-VirtualBox systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Jul 03 19:53:36 yashdeep-kaur-VirtualBox systemd[1]: Started vsftpd.service - vsftpd FTP server.
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$
```

Figure 4.1 - vsftpd FTP service status - active (running).



```
#
# Uncomment this to allow local users to upload files to local user's home directory
local_enable=YES
#
# Uncomment this to enable anonymous access to the FTP site
write_enable=YES
#
# Default umask for local users is 070.  This value determines the
# permissions of the files and directories that are created.
```

Figure 4.2 - local_enable and write_enable uncommented in vsftpd.conf.

4.2 File Transfer Test

```
echo "This is a test file for FTP transfer" > testfile.txt ftp
10.0.2.15      Name: developer1      Password: *****

ftp> put testfile.txt
ftp> get testfile.txt testfile_downloaded.txt
ftp> bye

ls -l testfile.txt testfile_downloaded.txt
```



```
diff testfile.txt testfile_downloaded.txt
```

```
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ftp 10.0.2.15
Connected to 10.0.2.15.
220 (vsFTPd 3.0.5)
Name (10.0.2.15:yashdeep-kaur): developer1
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Figure 4.3 - Successful FTP login as developer1.

```
Using binary mode to transfer files.
ftp> put testfile.txt
local: testfile.txt remote: testfile.txt
229 Entering Extended Passive Mode (|||8470|)
150 Ok to send data.
100% |*****| 51 664.06 KiB/s 00:00 ETA
226 Transfer complete.
51 bytes sent in 00:00 (197.63 KiB/s)
ftp>
```

Figure 4.4 - File upload (put) - transfer complete, 51 bytes sent.

```
ftp> get testfile.txt testfile_downloaded.txt
local: testfile_downloaded.txt remote: testfile.txt
229 Entering Extended Passive Mode (|||36584|)
150 Opening BINARY mode data connection for testfile.txt (51 bytes).
100% |*****| 51 264.91 KiB/s 00:00 ETA
226 Transfer complete.
51 bytes received in 00:00 (65.44 KiB/s)
ftp>
```

Figure 4.5 - File download (get) - transfer complete, 51 bytes received.

```
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ls -l testfile.txt testfile_downloaded.txt
-rw-rw-r-- 1 yashdeep-kaur yashdeep-kaur 51 Jul 3 20:07 testfile.txt
-rw-rw-r-- 1 yashdeep-kaur yashdeep-kaur 51 Jul 4 20:26 testfile_downloaded.txt
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ diff testfile.txt testfile_downloaded.txt
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$
```

Figure 4.6 - File integrity verification - both files 51 bytes, diff produced no output.

Task 5: Web Server Deployment

Objective: Install and configure Apache to host a webpage displaying project and student details.

5.1 Commands Executed

```
sudo apt install apache2 -y
sudo systemctl enable apache2
sudo systemctl start apache2
sudo systemctl status apache2
sudo nano /var/www/html/index.html
```

```
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 20:16:32 IST; 49s ago
     Invocation: efa1487ddc684bf692f36fd38cfdff3b
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 6762 (apache2)
      Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
        Tasks: 55 (limit: 1917)
      Memory: 5.4M (peak: 5.7M)
         CPU: 199ms
    CGroup: /system.slice/apache2.service
            └─6762 /usr/sbin/apache2 -k start -DFOREGROUND
              └─6778 /usr/sbin/apache2 -k start -DFOREGROUND
                └─6779 /usr/sbin/apache2 -k start -DFOREGROUND

Jul 03 20:16:31 yashdeep-kaur-VirtualBox systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 03 20:16:32 yashdeep-kaur-VirtualBox apache2[6762]: AH00558: apache2: Could not reliably determine the server's f
Jul 03 20:16:32 yashdeep-kaur-VirtualBox systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

Figure 5.1 - Apache2 service status - active (running).

5.2 Webpage Content (index.html)

```
<!DOCTYPE html>

<html>

<head><title>ABC Tech Solutions</title></head>

<body style="font-family:Arial; text-align:center; margin-top:50px;">

  <h1>Build and Configure a secure Linux server for a startup
company</h1>

  <p><strong>Name:</strong> Yashdeep Kaur</p>

  <p><strong>Roll Number:</strong> 2302711</p>

  <p><strong>Branch:</strong> Computer Science & Engineering</p>

  <p><strong>Current date & Time:</strong> <span
id="datetime"></span></p>

  <h3>Welcome to ABC Tech Solutions Pvt. Ltd. server!</h3>

  <script>      document.getElementById("datetime").innerHTML = new
Date().toLocaleString();  </script>

</body>

</html>
```

5.3 Result

Build and Configure a secure Linux server for a startup company

Name: Yashdeep kaur
Roll Number:2302711
branch:Computer Science & Engineering
Current date & Time:7/4/2026, 12:21:46 PM
Welcome to ABC Tech Solutions Pvt. Ltd. server!

Figure 5.2 - Hosted webpage displayed in browser with live date and time.

Task 6: Bash Automation Script - server_health.sh

Objective: Develop a Bash script that displays key server health metrics in a well-formatted manner.

6.1 Script Listing

```
#!/bin/bash
echo "=====
echo "          SERVER HEALTH REPORT"
echo "=====
echo "Current Date & Time : $(date)"
echo "Logged-in User      : $(whoami)"
echo "Hostname            : $(hostname)"
echo "IP Address           : $(hostname -I)"
echo
echo "===== Disk Usage ====="
df -h /
echo
echo "===== Memory Usage ====="
free -h
echo
echo "===== CPU Load ====="
uptime
echo
echo "===== System Uptime ====="
uptime -p
echo "=====
```

6.2 Execution

```
chmod +x server_health.sh ./server_health.sh
```

```

yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ chmod +x server_health.sh
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ./server_health.sh
=====
SERVER HEALTH REPORT
=====
Current Date & Time : Sat Jul 4 11:58:39 AM IST 2026
Logged-in User      : yashdeep-kaur
Hostname            : yashdeep-kaur-VirtualBox
IP Address          : 10.0.2.15 fd17:625c:f037:2:a00:27ff:fe7b:4695

=====DISK USAGE=====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2       25G   12G   12G   51% /

=====MEMORY USAGE=====
              total    used       free   shared  buff/cache   available
Mem:          1.6Gi     885Mi       64Mi     33Mi     871Mi       756Mi
Swap:         3.8Gi     136Mi       3.7Gi

=====CPU LOAD=====
11:58:39 up 12 min,  1 user,  load average: 0.06, 0.42, 0.42

=====SYSTEM UPTIME=====
up 12 minutes
=====

```

Figure 6.1 - server_health.sh output - date/time, user, hostname, IP, disk, memory, CPU load, uptime.

Bonus Challenge: Menu-Driven Linux Admin Tool

Objective: Build a menu-driven Bash application providing common administrative operations, running continuously until the user chooses to exit.

B.1 Script Listing (linux_admin_tool.sh)

```

#!/bin/bash
BACKUP_DIR="/company/backups"
while true;
do
echo ""
echo "1. Create User  2. Delete User  3. Disk Usage"
echo "4. Memory Usage 5. System Info  6. Logged-in Users"
echo "7. Create Backup Dir  8. Exit"
read -rp "Choose option [1-8]: " ch
case $ch in
    1) read -rp "Username: " u;
    sudo useradd -m "$u" && sudo passwd "$u" ;;
    2) read -rp "Username: " u; sudo userdel -r "$u" ;;
    3) df -h ;;
    4) free -h ;;
    5) echo "Host: $(hostname)"

```

```

Uptime: $(uptime -p)
IP: $(hostname -I)" ;;
6) who ;;
7) sudo mkdir -p "$BACKUP_DIR";
sudo chmod 700 "$BACKUP_DIR";
echo "Backup dir created" ;;
8) echo "Bye!";
exit 0 ;;
*) echo "Invalid option" ;;
esac
done

```

B.2 Execution

```

yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ./linux_admin_tool.sh

1.Create User 2.Delete User 3.Disk Usage
4.Memory Usage 5.System Info 6.Logged-In users
7.Create Backup Dir 8.Exit
Choose Option[1-8]: 3
Host: yashdeep-kaur-VirtualBox Uptime: up 29 minutes IP: 10.0.2.15 fd17:625c:f037:2:a00:27ff:fe7b:4695

1.Create User 2.Delete User 3.Disk Usage
4.Memory Usage 5.System Info 6.Logged-In users
7.Create Backup Dir 8.Exit
Choose Option[1-8]: 3
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            329M  2.3M  327M   1% /run
/dev/sda2        25G   12G   12G  51% /
tmpfs            822M    0  822M   0% /dev/shm
none             1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
none             1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs            822M  8.0K  822M   1% /tmp
tmpfs            165M   72K  165M   1% /run/user/1000

1.Create User 2.Delete User 3.Disk Usage
4.Memory Usage 5.System Info 6.Logged-In users
7.Create Backup Dir 8.Exit
Choose Option[1-8]: 7
[sudo: authenticate] Password:
BACKUP DIR created

1.Create User 2.Delete User 3.Disk Usage
4.Memory Usage 5.System Info 6.Logged-In users
7.Create Backup Dir 8.Exit
Choose Option[1-8]: 8
BYE!
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$

```

Figure B.1 - linux_admin_tool.sh menu execution - System Info, Disk Usage, Backup Dir creation, and Exit.

```

yashdeep-kaur@yashdeep-kaur-VirtualBox:~$ ls -ld /company/backups
drwx----- 2 root root 4096 Jul 3 18:05 /company/backups
yashdeep-kaur@yashdeep-kaur-VirtualBox:~$

```

Figure B.2 - Backup directory verified with correct root-only (700) permissions.

Challenges Encountered

- A typo while creating the /company directory structure resulted in a folder named 'developer' instead of 'developers', causing the chgrp command to fail with 'No such file or directory'. This was identified by listing directory contents and corrected using mv.
- Apache displayed a startup warning (AH00558) about not being able to reliably determine the server's fully qualified domain name. This is a cosmetic warning only and did not affect functionality.
- The first version of server_health.sh failed with 'hostname-I: command not found' because a required space between hostname and -I was accidentally omitted while typing. This was corrected to hostname -I.
- The hosted webpage initially did not display the live date and time. Inspecting the HTML source revealed a typo in the JavaScript method name (toLocateString instead of toLocaleString), which caused the script to fail silently. Correcting the method name resolved the issue.
- Only a single VM was available for this assignment, so the SSH and FTP demonstrations were performed using the server's own network IP address (10.0.2.15) rather than a separate physical client machine.

Learning Outcomes and Conclusion

Learning Outcomes

- Gained hands-on experience configuring Linux users, groups, and permission structures for a multi-team organizational environment.
- Understood how Linux file permissions (rwx, chmod, chgrp) enforce access control between different user groups.
- Learned to configure and verify SSH for secure remote administration.
- Learned to configure vsftpd for FTP-based file transfer and verify data integrity using diff.
- Deployed and hosted a live website using the Apache2 web server.
- Developed Bash scripts to automate routine administrative and monitoring tasks, including a menu-driven interactive tool.
- Practiced systematic debugging of real configuration and scripting errors, mirroring real-world DevOps troubleshooting.

Conclusion

This assignment provided practical, end-to-end experience in setting up a production-style Linux server for a simulated startup company. All required tasks - user and group management, directory permissions, SSH, FTP, Apache web hosting, and Bash automation - were completed successfully, along with the optional bonus challenge of building a menu-driven administration tool. The exercise reinforced core Linux system administration skills that form the foundation of entry-level DevOps and Cloud Computing roles.